

Digital Twin for Surface Treatment Industries – Detailed Report

1. Introduction

This project demonstrates a Digital Twin (DT) system for surface treatment industries. It integrates Discrete Event Simulation (DES) and optimization methods to support finite capacity production planning. The system aims to improve planning efficiency, decision-making, and resource utilization in surface treatment workshops.

Surface treatment industries face challenges such as high product variety, short lead times, multi-stage processes, and capacity constraints. A digital twin allows simulation of production operations and optimization of scheduling and batching, bridging the gap between planning and execution.

2. Project Objectives

The main objectives of this project are:

1. Develop a simulation framework for surface treatment workshops using DES.
2. Implement optimization services for scheduling, batch grouping, and finite capacity planning.
3. Provide decision support tools, including timeline visualization and performance metrics.
4. Demonstrate a mini-project scenario for single and multi-workshop operations.
5. Lay the foundation for future extensions, such as reinforcement learning and real-time data integration.

3. Digital Twin Architecture

The Digital Twin system includes two main components:

3.1 Physical System:

- Represents surface treatment workshops with multiple stations.
- Includes machines, operators, and material handling processes.
- Models the actual flow of orders and multi-stage processes.

3.2 Cyber System:

- Simulation Module: Implements DES models using SimPy.
- Optimization Module: Provides scheduling and batch grouping algorithms.
- Visualization Module: Creates dashboards and performance charts.
- Enables bidirectional interaction: planning decisions affect production execution, and production outcomes inform planning.

4. Finite Capacity Planning

Finite capacity planning ensures that production respects resource limits:

- Multiple workshops are coordinated with capacity constraints.
- Orders are batched and sequenced to minimize lead time and resource idle time.
- Simulation models allow testing different scheduling strategies.
- Optimization helps maximize throughput and operator utilization.

5. Methodology

The project methodology combines simulation and optimization:

1. Simulation (DES):

- SimPy-based models simulate workshops, orders, and resources.
- Captures stochastic processing times, machine breakdowns, and operator availability.

2. Optimization:

- Scheduling algorithms determine order sequences and batch groups.
- Batch grouping ensures efficient machine usage and minimal lead time.
- Multi-objective optimization considers throughput, WIP, and operator versatility.

3. Visualization:

- Dashboards display timelines, performance metrics, and resource utilization.
- Allows planners to observe workflow bottlenecks and evaluate scenarios.

4. Case Studies:

- Simple single-workshop scenario.
- Multi-workshop scenario with resource coordination and batch sequencing.

6. Technologies Used

Component	Technology / Tool
Simulation	Python, SimPy
Optimization	NumPy, SciPy
Data Analysis	Pandas
Visualization	Matplotlib, Plotly
Language	Python 3.8+

7. Results and Discussion

- The simulation accurately reproduces workshop workflows.
- Scheduling and batching optimization reduce lead times and improve resource usage.
- Dashboards provide visual feedback on order progress and performance metrics.
- Multi-workshop coordination demonstrates scalable planning for complex scenarios.

This mini-project validates that digital twin technology can improve short-term production planning in surface treatment industries.

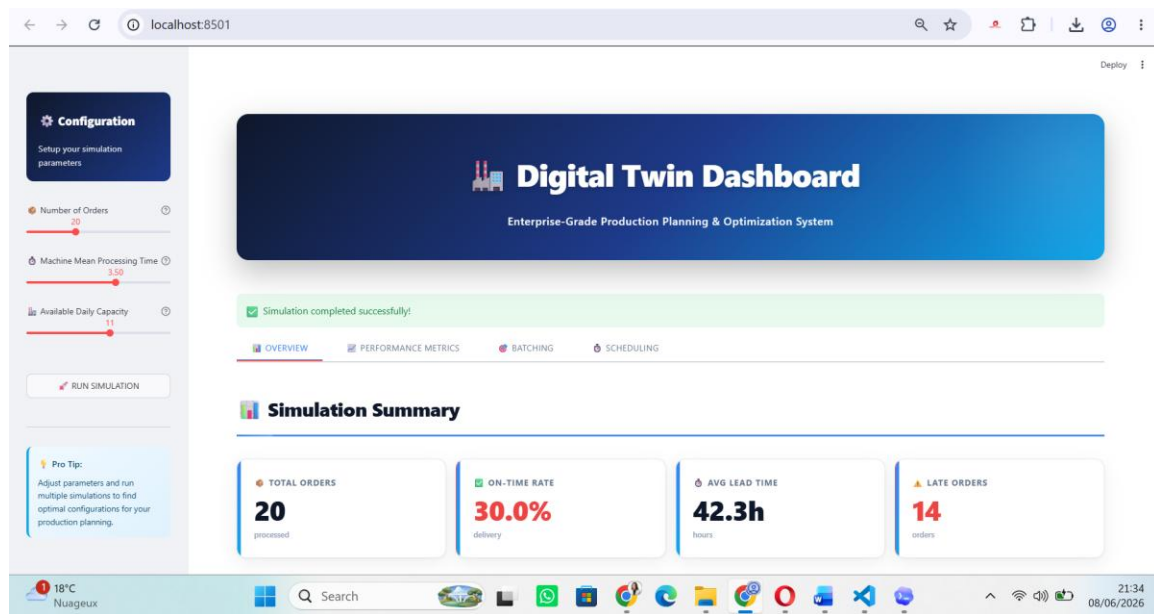
8. Future Extensions

- Integration of Reinforcement Learning for adaptive scheduling.
- Real-time data collection from workshop sensors.
- Database integration for historical order and performance tracking.
- Web dashboards for planner interaction and decision support.
- Advanced optimization algorithms, e.g., genetic algorithms, for better batch sequencing.

9. Practical implementation

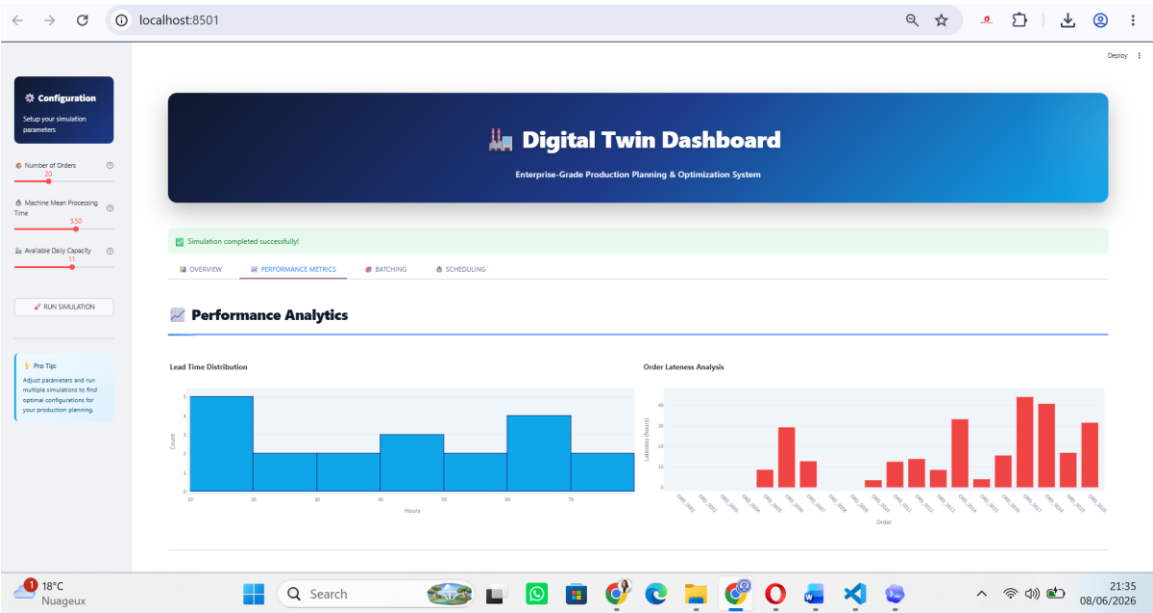
Here is practical implementation screenshots of this project:

Screenshot 1: Simulation Overview



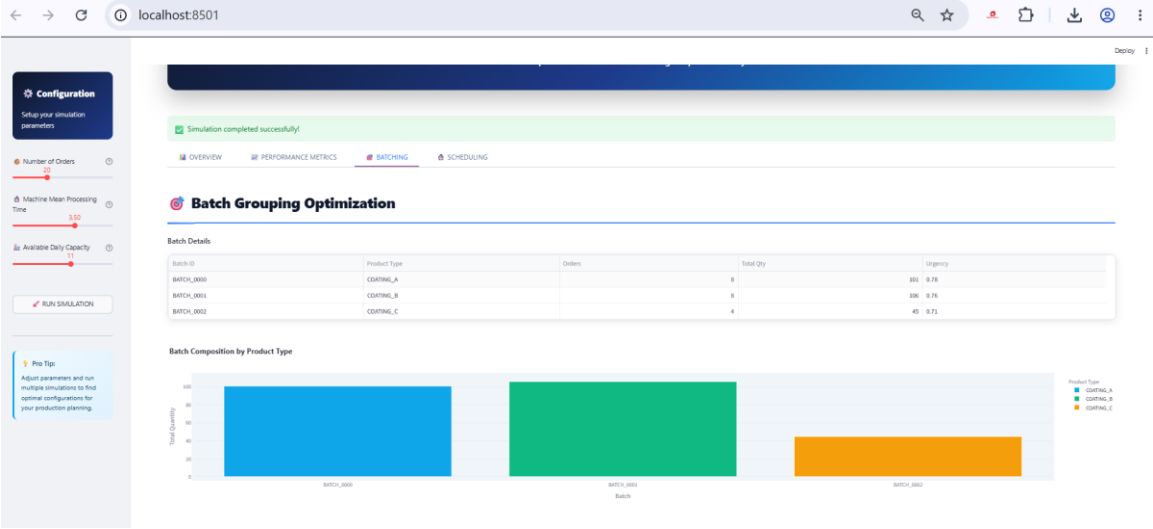
This screen displays the **Digital Twin Dashboard Overview**. It shows the **total orders processed (20)**, the **on-time delivery rate (30%)**, **average lead time (42.3 hours)**, and **late orders (14)**. This view helps understand the **overall performance** of the simulated production workflow and highlights bottlenecks in order proce

Screenshot 2: Performance Metrics



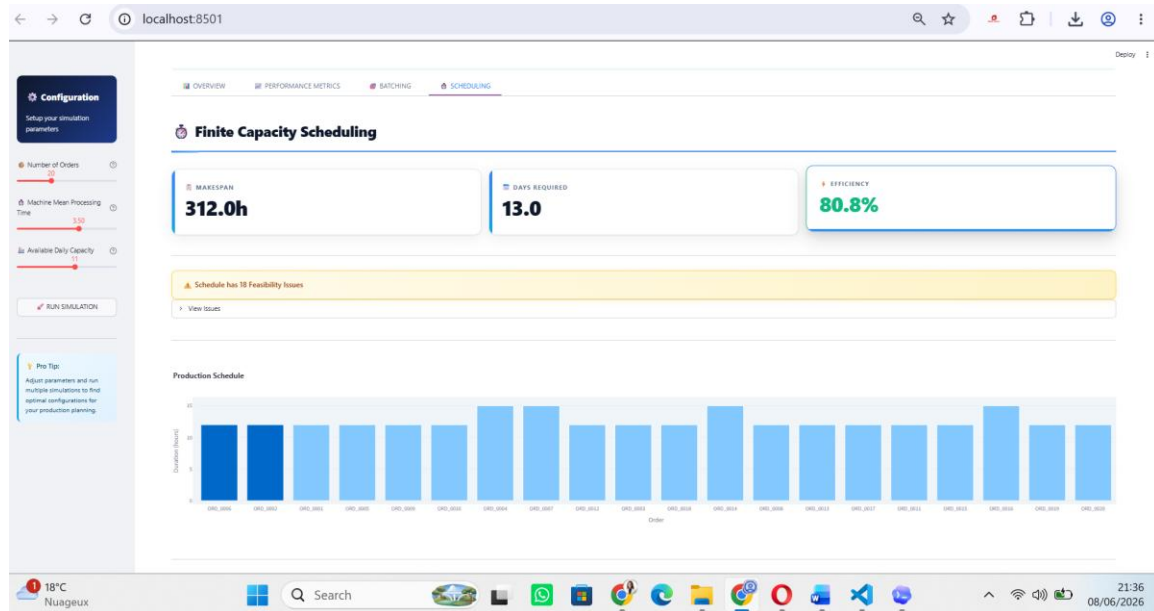
This screen shows **performance analytics**, including **Lead Time Distribution** and **Order Lateness Analysis**. The charts visualize how long orders take to complete and which orders are delayed. This allows planners to **identify variability in processing times** and focus on improving scheduling efficiency.

Screenshot 3: Batch Grouping Optimization



This screen shows **batch optimization results**. It lists batches with **product types, order counts, and urgency levels**, and provides a chart of **batch composition by product type**. This view helps planners **group orders efficiently**, reducing machine idle time and improving resource utilization.

Screenshot 4: Finite Capacity Scheduling



This screen displays **finite capacity scheduling results**, including **total makespan (312.0 h)**, **days required (13)**, and **efficiency (80.8%)**. The production schedule chart shows order sequencing across machines. This is critical for **evaluating workflow feasibility** under resource constraints and planning realistic production timelines.

10. Conclusion

This project demonstrates a Digital Twin system for surface treatment industries. By combining DES and optimization, it provides planners with decision support, improves resource utilization, and supports multi-workshop coordination. The framework is modular and can be extended for real-time adaptive planning, making it suitable for industrial applications and as a foundation for PhD-level research.

Summary:

Digital Twin for Surface Treatment Industries

A mini project demonstrating Discrete Event Simulation (DES) + Optimization coupling for finite capacity production planning in surface treatment industries.



1. INTRODUCTION

This project implements a digital twin system for short-term production planning in surface treatment industries. It combines Discrete Event Simulation (DES) with optimization services to support scheduling, batch grouping and capacity planning under finite resource constraints. The project provides planners with actionable insights through dashboards and performance metrics.

2. INDUSTRY CONTEXT

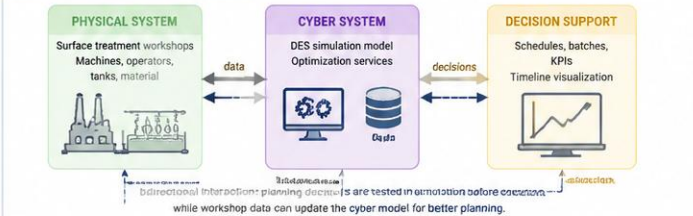
- Made-to-order production
- Short promised lead times
- High customer diversity
- Highly variable product mix
- Mix of automated and manual processes
- Need for capacity agility
- Complex batch compositions
- Multi-workshop coordination



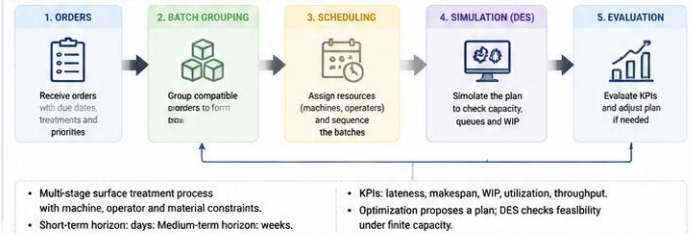
3. PROJECT OBJECTIVES

1. Build a DES model of surface treatment workshops using SimPy.
2. Develop optimization services for scheduling and batch grouping.
3. Plan production under finite capacity constraints.
4. Provide decision support through visualization and KPIs.
5. Create a modular framework for future extensions.

4. DIGITAL TWIN ARCHITECTURE



5. FINITE CAPACITY PLANNING METHODOLOGY



6. IMPLEMENTATION AND TECHNOLOGIES

6.1 PROJECT STRUCTURE



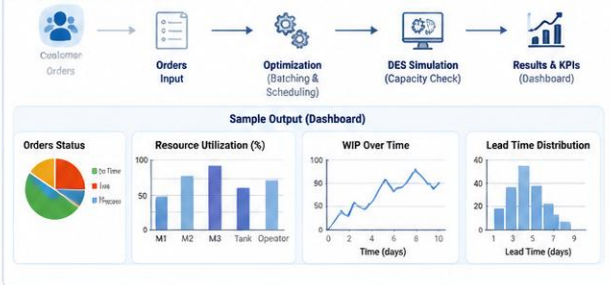
6.2 TECHNOLOGY STACK

Component	Technology
Simulation	SimPy
Optimization	NumPy / SciPy
Data Analysis	Pandas
Visualization	Matplotlib, Plotly
Configuration	YAML
Language	Python 3.8+

What is Included

- ✓ SimPy-based simulation framework
- ✓ Workshop model (single and multi-stage)
- ✓ Order and batch management
- ✓ Optimization skeleton
- ✓ Example scenarios
- ✓ Performance metrics and dashboards

7. EXAMPLE WORKFLOW



8. KEY BENEFITS



9. FUTURE EXTENSIONS

- Reinforcement learning for adaptive scheduling
- Real-time data collection interface
- Database integration for order history
- Web dashboard for planners and supervisors
- Advanced optimization (GA, Tabu Search, etc.)
- What-if analysis and scenario comparison



10. CONCLUSION

This mini project demonstrates how simulation, optimization and decision support are integrated in a digital twin framework to address finite capacity planning challenges in surface treatment industries. It provides a strong foundation for further research and development in the context of short-term production planning.